

KALINGA INSTITUTE OF INDUSTRIAL TECHNOLOGY

Deemed to be University, Bhubaneswar - 751024, Odisha

MID SEMESTER EXAMINATION — SOLUTIONS & MARKING SCHEME

Subject: DATABASE MANAGEMENT SYSTEMS

Time: 1.5 Hours

Semester / Programme: 4

Full Marks: 20

Instructions: Answer Question No. 1 (Compulsory) and any THREE from Question No. 2 to 5. Figures in brackets indicate marks.

SECTION - A (Compulsory)

1(a) What is the primary difference between a physical schema and a logical schema?

[1]

Solution & Key Concept:

- Physical schema describes how the data is stored on storage devices.
- Logical schema describes what data is stored and the relationships among them.

Answer: Physical schema defines storage details (e.g., file structures), while logical schema defines the database structure and constraints.

Marking Rubric: [1 Mark] for clear distinction between physical and logical abstraction levels

1(b) Define the term 'super key' in the relational model.

[1]

Solution & Key Concept:

- A super key is a set of attributes that uniquely identifies a tuple in a relation.
- No two distinct tuples can have the same value for all attributes in the super key.

Answer: A set of one or more attributes that, taken collectively, allows us to uniquely identify a tuple in the relation.

Marking Rubric: [1 Mark] for correct definition

1(c) Express the Armstrong's Axiom of Transitivity in symbolic notation.

[1]

Formula: If $X \twoheadrightarrow Y$ and $Y \twoheadrightarrow Z$, then $X \twoheadrightarrow Z$

Solution & Key Concept:

- If X determines Y and Y determines Z, then X determines Z.

Answer: If $X \twoheadrightarrow Y$ and $Y \twoheadrightarrow Z$, then $X \twoheadrightarrow Z$.

Marking Rubric: [1 Mark] for correct notation

1(d) What is the purpose of the 'NULL' value in SQL?

[1]

Solution & Key Concept:

- NULL represents an unknown, missing, or inapplicable value in a database attribute.

Answer: It represents the absence of a value or an unknown state.

Marking Rubric: [1 Mark] for indicating unknown or missing data

1(e) Identify the primary advantage of using a DBMS over a file processing system.

[1]

Solution & Key Concept:

- DBMS provides data independence, reduced redundancy, and centralized control.

Answer: Data independence and reduced redundancy.

Marking Rubric: [1 Mark] for mentioning one of the core benefits

SECTION - B (Answer any THREE)

Answer any THREE questions from Q.2 to Q.5. Each question carries 5 marks.

Q.2. (a) Explain the three-schema architecture of a database system. [2.5]

Solution for (a):

- 1. Internal Level: Describes physical storage.
- 2. Conceptual Level: Describes data and relationships.
- 3. External Level: Describes user views.

Result / Conclusion: The architecture separates data into three levels to achieve data independence.

Marking Rubric: [1.5 Marks] for naming all three, [1 Mark] for explaining their purpose

(b) Differentiate between Total Participation and Partial Participation in E-R modeling with an example. [2.5]

Solution for (b):

- Total participation: Every entity in the set must be involved in the relationship.
- Partial participation: Some entities may not be involved.

Result / Conclusion: Example: Employee works for Department (Total if every employee must belong to a department).

Marking Rubric: [1.5 Marks] for definitions, [1 Mark] for valid example

Q.3. (a) Given the relation $R(A, B, C, D)$ and functional dependencies $F = \{A \twoheadrightarrow B, B \twoheadrightarrow C, C \twoheadrightarrow D\}$. Find the closure of A (A^+). [2]

A^+

Solution for (a):

- Start with $A^+ = \{A\}$
- Use $A \twoheadrightarrow B$: $A^+ = \{A, B\}$
- Use $B \twoheadrightarrow C$: $A^+ = \{A, B, C\}$
- Use $C \twoheadrightarrow D$: $A^+ = \{A, B, C, D\}$

Result / Conclusion: $A^+ = \{A, B, C, D\}$

Marking Rubric: [1 Mark] for showing steps, [1 Mark] for final result

(b) State and prove the Augmentation rule of Armstrong's Axioms. [3]

Formula: If $X \twoheadrightarrow Y$, then $XZ \twoheadrightarrow YZ$

Solution for (b):

- Definition: If $X \twoheadrightarrow Y$, then for any attribute set Z, $XZ \twoheadrightarrow YZ$.
- Proof: By reflexivity, $Z \twoheadrightarrow Z$. Given $X \twoheadrightarrow Y$, then $XZ \twoheadrightarrow YZ$ using transitivity.

Result / Conclusion: Proven by combining reflexivity and transitivity.

Marking Rubric: [1 Mark] for definition, [2 Marks] for logical proof

Q.4. (a) Explain the concept of 'Lossless Decomposition' in database normalization. [2.5]
Formula: $R_1 \bowtie R_2 \mapsto R_1$ or $R_1 \cap R_2 \mapsto R_2$

Solution for (a):

- Decomposition is lossless if the join of the decomposed relations reproduces the original relation without extra tuples (spurious tuples).

Result / Conclusion: Ensures no information is lost and no invalid information is generated during join.

Marking Rubric: [1.5 Marks] for concept, [1 Mark] for formula/condition

(b) Briefly define 1NF and 2NF. [2.5]

Solution for (b):

- 1NF: Attributes must be atomic (no multi-valued).
- 2NF: Must be in 1NF and no partial dependencies.

Result / Conclusion: 1NF requires atomicity; 2NF requires 1NF plus the removal of partial functional dependencies.

Marking Rubric: [1.25 Marks] for each definition

Q.5. (a) Write a SQL query to find the names of employees who earn more than the average salary of their respective departments. [3]

Solution for (a):

- Use a correlated subquery.
- `SELECT name FROM Emp E1 WHERE salary > (SELECT AVG(salary) FROM Emp E2 WHERE E1.dept_id = E2.dept_id)`

Result / Conclusion: `SELECT name FROM Emp E1 WHERE salary > (SELECT AVG(salary) FROM Emp E2 WHERE E1.dept_id = E2.dept_id);`

Marking Rubric: [1 Mark] for outer query, [2 Marks] for correlated logic

(b) Explain the difference between 'WHERE' and 'HAVING' clauses in SQL. [2]

Solution for (b):

- WHERE filters rows before grouping.
- HAVING filters groups after the aggregate function is applied.

Result / Conclusion: WHERE acts on individual records; HAVING acts on grouped data.

Marking Rubric: [1 Mark] for WHERE, [1 Mark] for HAVING

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